COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating System Management	**SECTION: CPE31S3**	
DATE PERFORMED:	9/5/2026	**DATE SUBMITTED:**	9/5/2026
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ACTIVITY NO. 6.1	Targeting Specific Nodes

1. PROCEDURE OUTPUT

1. Create a new playbook and named it site.yml. Follow the commands as shown in the image below. Make sure to save the file and exit.

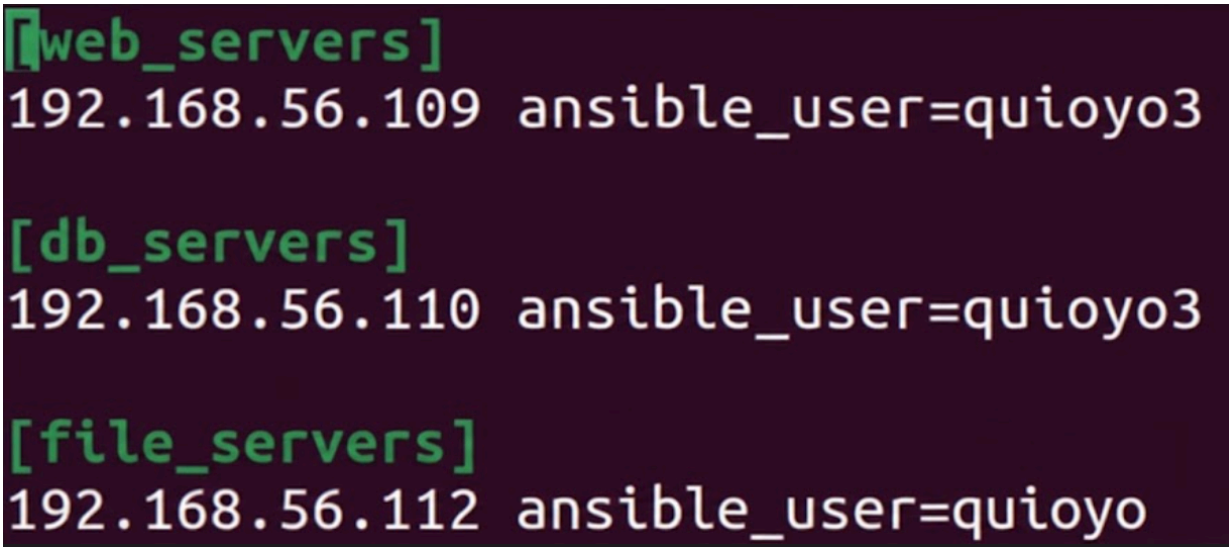
```
- hosts: all
  become: true
  tasks:

  - name: install apache and php for Ubuntu servers
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

    - name: install apache and php for CentOS servers
      dnf:
        name:
          - httpd
          - php
        state: latest
      when: ansible_distribution == "CentOS"
```

2. Edit the inventory file. Remove the variables we put in our last activity and group according to the image shown below:
Make sure to save the file and exit.
Right now, we have created groups in our inventory file and put each server in its own group. In

other cases, you can have a server be a member of multiple groups, for example you have a test server that is also a web server.



3. Edit the site.yml by following the image below:
Make sure to save the file and exit.
The pre-tasks command tells the ansible to run it before any other thing. In the pre-tasks, CentOS will install updates while Ubuntu will upgrade its distribution package. This will run before running the second play, which is targeted at web_servers. In the second play, apache and php will be installed on both Ubuntu servers and CentOS servers.
Run the site.yml file and describe the result.

```
- hosts: all
  become: true
  pre_tasks:

  - name: install updates (CentOS)
    dnf:
      update_only: yes
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install updates (Ubuntu)
    apt:
      upgrade: dist
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

- hosts: web_servers
  become: true
  tasks:

  - name: install apache and php for Ubuntu servers
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
      update_cache: yes
```

4. Let's try to edit again the site.yml file. This time, we are going to add plays targeting the other servers. This time we target the db_servers by adding it on the current site.yml. Below is an example: (Note add this at the end of the playbooks from task 1.3.
Make sure to save the file and exit.
Run the site.yml file and describe the result.

5. Go to the remote server (Ubuntu) terminal that belongs to the db_servers group and check the status for mariadb installation using the command: systemctl status mariadb. Do this on the CentOS server also.
Describe the output.

```
ok: [192.168.56.12]

TASK [install updates (CentOS)] ***********************************************************************************************
skipping: [192.168.56.9]
skipping: [192.168.56.10]
ok: [192.168.56.12]

TASK [install updates (Ubuntu)] ***********************************************************************************************
skipping: [192.168.56.12]
ok: [192.168.56.9]
ok: [192.168.56.10]

PLAY [web_servers] ************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************
ok: [192.168.56.9]

TASK [install apache and php for Ubuntu servers] ******************************************************************************
ok: [192.168.56.9]

TASK [install apache and php for CentOS servers] ******************************************************************************
skipping: [192.168.56.9]

PLAY [db_servers] *************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************
ok: [192.168.56.10]

TASK [install mariadb package (CentOS)] ***************************************************************************************
skipping: [192.168.56.10]

TASK [Mariadb- Restarting/Enabling] *******************************************************************************************
fatal: [192.168.56.10]: FAILED! => {"changed": false, "msg": "Could not find the requested service mariadb: host"}

PLAY RECAP ********************************************************************************************************************
192.168.56.10              : ok=3    changed=0    unreachable=0    failed=1    skipped=2    rescued=0    ignored=0
192.168.56.12              : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

6. Edit the site.yml again. This time we will append the code to configure installation on the file_servers group. We can add the following on our file.
Make sure to save the file and exit.
Run the site.yml file and describe the result.
The testing of the file_servers is beyond the scope of this activity, and as well as our topics and objectives. However, in this activity we were able to show that we can target hosts or servers using grouping in ansible playbooks.

```
mariadb.service - MariaDB 10.6.23 database server
     Loaded: loaded (/lib/systemd/system/mariadb.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-09-03 09:48:24 CST; 26s ago
       Docs: man.mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 36857 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /ar/mysld (code=exited, statu
    Process: 36851 ExecStartPre=/bin/sh -c systemctl unset-environment _WSREP_STOSTION (code=exited, stat
    Process: 36801 ExecStartPreR=/bin/sh c [ ! -e /usr/bin/galera_recov && VAR= ||  VAR='/usr/bin/galera_
    Process: 36903 ExecStartPost=/etc/mysql/debian-start (code=exited, _WSREP_STATSTION (code=exited, sta
   Main PID: 36890 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 12 (limit: 30342)
     Memory: 61.3M
        CPU: 767ms
     CGroup: /system.slice/mariadb.service
             └─36890 /usr/sbin/mariadbd
```

Task 2: Using Tags in running playbooks
In this task, our goal is to add metadata to our plays so that we can only run the plays that we want to run, and not all the plays in our playbook.

1. Edit the site.yml file. Add tags to the playbook. After the name, we can place the tags: name_of_tag. This is an arbitrary command, which means you can use any name for a tag.
Make sure to save the file and exit.
Run the site.yml file and describe the result.

```
PLAY [web_servers] ***************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************
ok: [192.168.56.10]

TASK [install apache and php for Ubuntu servers] *********************************************************************
ok: [192.168.56.10]

TASK [install apache and php for CentOS servers] *********************************************************************
skipping: [192.168.56.10]

PLAY [db_servers] ****************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************
ok: [192.168.56.9]

TASK [install mariadb package (CentOS)] ******************************************************************************
skipping: [192.168.56.9]

TASK [install mariadb package (Ubuntu)] ******************************************************************************
ok: [192.168.56.9]

TASK [Mariadb- Restarting/Enabling] **********************************************************************************
changed: [192.168.56.9]

PLAY [file_servers] **************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************
ok: [192.168.56.12]

TASK [install samba package] *****************************************************************************************
ok: [192.168.56.12]

PLAY RECAP ***********************************************************************************************************
192.168.56.10              : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.12              : ok=4    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=5    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

2. On the local machine, try to issue the following commands and describe each result:

2.1 ansible-playbook --list-tags site.yml

```
quioyo3@workshop:~/CPE212_quioyo$ ansible-playbook --list-tags site.yml

playbook: site.yml

  play #1 (all): all    TAGS: []
      TASK TAGS: [always]

  play #2 (web_servers): web_servers    TAGS: []
      TASK TAGS: [apache, apache2, centos, httpd, ubuntu]

  play #3 (db_servers): db_servers      TAGS: []
      TASK TAGS: [centos, db, mariadb, ubuntu]

  play #4 (file_servers): file_servers  TAGS: []
      TASK TAGS: [samba]
```

2.2 ansible-playbook --tags centos --ask-become-pass site.yml

```
TASK [install updates (CentOS)] ************************************************************************************
skipping: [192.168.56.10]
skipping: [192.168.56.9]
ok: [192.168.56.12]

TASK [install updates (Ubuntu)] ************************************************************************************
skipping: [192.168.56.12]
ok: [192.168.56.9]
ok: [192.168.56.10]

PLAY [web_servers] *************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.10]

TASK [install apache and php for CentOS servers] *******************************************************************
skipping: [192.168.56.10]

PLAY [db_servers] **************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.9]

TASK [install mariadb package (CentOS)] ****************************************************************************
skipping: [192.168.56.9]

PLAY [file_servers] ************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.12]

PLAY RECAP *********************************************************************************************************
192.168.56.10              : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.12              : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

2.3 ansible-playbook --tags db --ask-become-pass site.yml

```
TASK [install updates (CentOS)] ************************************************************************************
skipping: [192.168.56.10]
skipping: [192.168.56.9]
ok: [192.168.56.12]

TASK [install updates (Ubuntu)] ************************************************************************************
skipping: [192.168.56.12]
ok: [192.168.56.9]
ok: [192.168.56.10]

PLAY [web_servers] *************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.10]

PLAY [db_servers] **************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.9]

TASK [install mariadb package (CentOS)] ****************************************************************************
skipping: [192.168.56.9]

TASK [install mariadb package (Ubuntu)] ****************************************************************************
ok: [192.168.56.9]

PLAY [file_servers] ************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.12]

PLAY RECAP *********************************************************************************************************
192.168.56.10              : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.12              : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

2.4 ansible-playbook --tags apache --ask-become-pass site.yml

```
TASK [install updates (CentOS)] ************************************************************************************
skipping: [192.168.56.10]
skipping: [192.168.56.9]
ok: [192.168.56.12]

TASK [install updates (Ubuntu)] ************************************************************************************
skipping: [192.168.56.12]
ok: [192.168.56.9]
ok: [192.168.56.10]

PLAY [web_servers] *************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.10]

TASK [install apache and php for Ubuntu servers] *******************************************************************
ok: [192.168.56.10]

TASK [install apache and php for CentOS servers] *******************************************************************
skipping: [192.168.56.10]

PLAY [db_servers] **************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.9]

PLAY [file_servers] ************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.12]

PLAY RECAP *********************************************************************************************************
192.168.56.10              : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.12              : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

2.5 ansible-playbook --tags “apache,db” --ask-become-pass site.yml

Task 3: Managing Services

1. Edit the file site.yml and add a play that will automatically start the httpd on CentOS server.
Figure 3.1.1
Make sure to save the file and exit.
You would also notice from our previous activity that we already created a module that runs a service.
Figure 3.1.2
This is because in CentOS, installed packages’ services are not run automatically. Thus, we need to create the module to run it automatically.

2. To test it, before you run the saved playbook, go to the CentOS server and stop the currently running httpd using the command sudo systemctl stop httpd. When prompted, enter the sudo password. After that, open the browser and enter the CentOS server’s IP address. You should not be getting a display because we stopped the httpd service already.

3. Go to the local machine and this time, run the site.yml file. Then after running the file, go again to the CentOS server and enter its IP address on the browser. Describe the result.
To automatically enable the service every time we run the playbook, use the command enabled: true similar to Figure 7.1.2 and save the playbook.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Supplementary Activity:
1. Creating a Role for Web Servers

2. Creating a Role for Web Servers
3. Creating a Role for Database Servers

```
quioyo@server1:~/CPE212_quioyo$ ls roles
db  web
```

4. Combining Multiple Roles in a Playbook

To orchestrate multiple roles in a single playbook for different server groups.

Add tags to each role to allow selective execution, e.g., tags: web for webserver, tags: db for Database.

```
- name: install apache and php for Ubuntu servers
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
    update_cache: true
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: start httpd (CentOS)
  service:
    name: httpd
    state: started
    enabled: true
  when: ansible_distribution == "CentOS"
```

```yaml
---
- name: install mariadb package (CentOS)
  dnf:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: install mariadb package (Ubuntu)
  apt:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: Mariadb - Restarting/Enabling
  service:
    name: mariadb
    state: started
    enabled: true
```

```yaml
- hosts: all
  become: true
  pre_tasks:
    - name: install updates (CentOS)
      dnf:
        update_only: yes
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: install updates (Ubuntu)
      apt:
        upgrade: dist
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

- hosts: web_servers
  become: true
  roles:
    - role: web
      tags: web

- hosts: db_servers
  become: true
  roles:
    - role: db
      tags: db
```

Reflections:
Answer the following:
1. What is the importance of putting our remote servers into groups?
   - Server grouping lets you run tasks on multiple nodes at once. You target web servers, database clusters, or staging environments with a single command. Grouping eliminates manual, repetitive execution and prevents human error.
2. What is the importance of tags in playbooks?
   - Tags let you run or skip specific tasks without executing the whole playbook. You save time during testing and updates. A tag allows you to restart a single service in seconds instead of running a twenty-minute configuration pipeline.
3. Why do you think some services need to be managed automatically in playbooks?
   - Automated service management ensures consistent system states across different operating systems. Ansible modules detect the underlying platform and apply the correct commands. You eliminate configuration drift, speed up deployments, and reduce downtime.

Conclusion:
In conclusion,I learned how to focus on remote hosts by arranging the Ansible inventory around server roles and by putting playbooks into host-specific plays. I used task tags and the built-in service modules to run or skip tasks while I managed services such as HTTPD smoothly across different operating systems. At the end I organized the tasks into roles so that the Ansible project followed best practices, for clean and maintainable structure.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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